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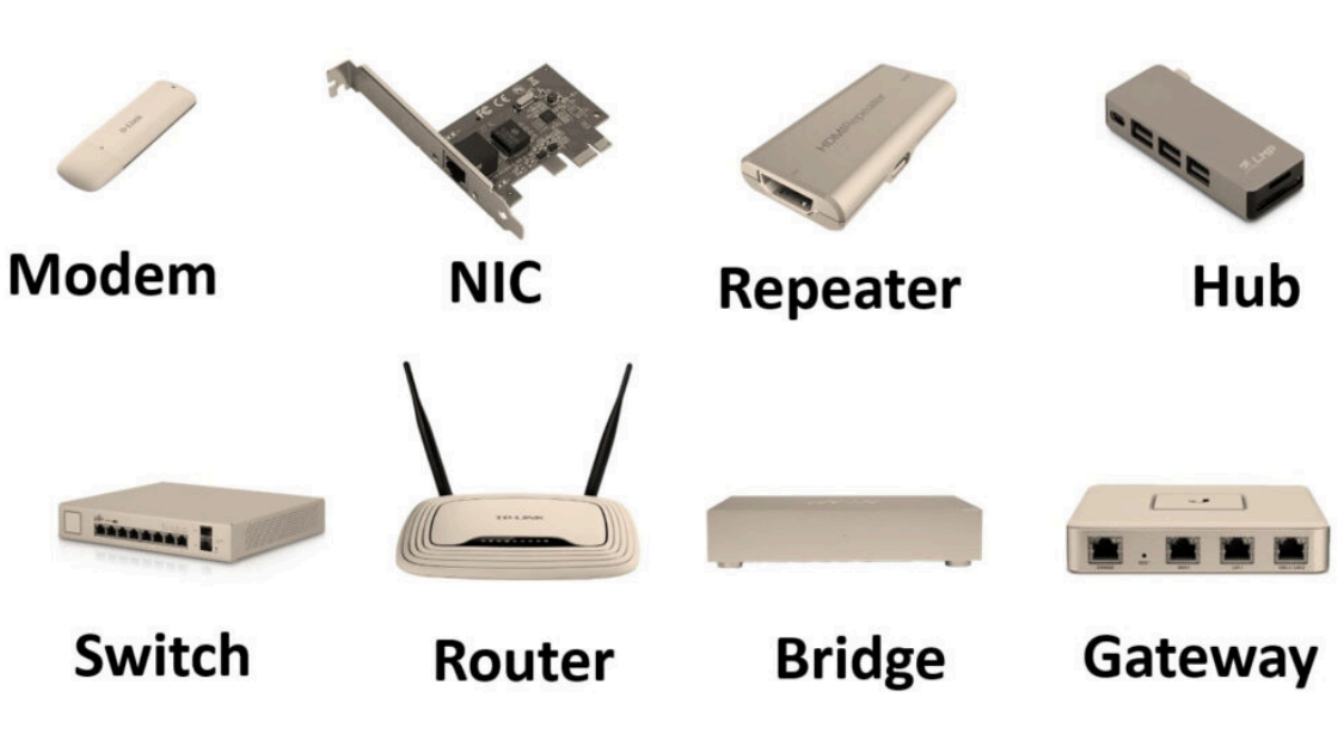
4. Basic networking devices



Types of Network Devices

1. Router

- **Function**: A router connects multiple networks and directs data packets between them. It acts as a gateway between your local network and the internet, deciding the best route for data to travel.
- **How It Works**: Routers use IP addresses to route packets to their destination. They analyze incoming data packets, determine the destination, and send the packets to the appropriate network.
- **Usage**: Essential for both home and office networks, allowing multiple devices to connect to the internet through a single connection.
- **Example**: Your home router connects all your devices (phones, laptops, smart TVs) to the internet. In larger networks, routers link different office branches.

2. Switch

- **Function**: A switch connects multiple devices within a single network, creating a local area network (LAN). It manages data packets at the data link layer and only forwards data to the intended device.
- **How It Works**: Switches use MAC (Media Access Control) addresses to send data to specific devices. When a data packet arrives, the switch checks the MAC address and sends it directly to the destination device, reducing unnecessary traffic.
- **Usage**: Used within networks to efficiently connect devices, especially in offices, schools, and data centers.
- **Example**: In an office LAN, a switch connects all computers to each other and to printers, enabling efficient internal communication.

3. Hub

- **Function**: A hub connects multiple devices in a network, but unlike a switch, it broadcasts incoming data to all connected devices. Hubs are simpler and cheaper than switches but less efficient.
- **How It Works**: Hubs don’t differentiate where data should go. They send the same data to all devices connected to them, creating more network traffic.
- **Usage**: Hubs are rarely used in modern networks due to inefficiency. They may still be found in simple setups but are largely replaced by switches.
- **Example**: If five computers are connected to a hub, a message from one computer is sent to all five devices, regardless of which one is the intended recipient.

4. Modem

- **Function**: A modem (Modulator-Demodulator) connects a home or business network to an internet service provider (ISP). It converts digital data from your devices into signals that can travel over phone lines, cable, or fiber.
- **How It Works**: Modems modulate (convert digital data to analog signals) for sending over long distances and demodulate (convert signals back to digital data) for receiving, making internet connections possible.
- **Usage**: Essential for internet connectivity; often combined with a router in a single device for home networks.
- **Example**: In DSL or cable internet, the modem connects your home network to your ISP.

5. Access Point (AP)

- **Function**: An access point allows wireless devices to connect to a wired network. It extends network coverage, especially in large areas or buildings.
- **How It Works**: Access points broadcast a Wi-Fi signal, enabling wireless devices to connect to the network without needing a physical Ethernet cable. They link wireless devices with the main network, typically using Ethernet.
- **Usage**: Widely used in offices, schools, and homes to provide Wi-Fi coverage.
- **Example**: A Wi-Fi router acts as both a router and an access point, allowing devices to connect wirelessly to the network.

6. Firewall

- **Function**: A firewall is a security device that monitors and controls incoming and outgoing network traffic based on predetermined security rules. It acts as a barrier between a trusted network and an untrusted network (e.g., the internet).
- **How It Works**: Firewalls inspect packets of data and determine whether to allow or block them based on set security rules. They can filter traffic by IP addresses, port numbers, and protocols.
- **Usage**: Essential for network security, protecting against unauthorized access, malware, and cyberattacks.
- **Example**: Firewalls are used in both homes and businesses to protect devices from internet threats. Many routers include built-in firewalls for added security.

7. Network Interface Card (NIC)

- **Function**: A NIC is a hardware component that allows a device to connect to a network, either via Ethernet (wired) or Wi-Fi (wireless).
- **How It Works**: NICs provide a unique MAC address for the device, enabling it to communicate on the network. They process data received from the network and prepare outgoing data for transmission.
- **Usage**: NICs are built into most modern devices but can also be added as an external component to connect non-networked devices.
- **Example**: A desktop computer has an Ethernet NIC for wired connections and may have a Wi-Fi NIC for wireless.

8. Repeater

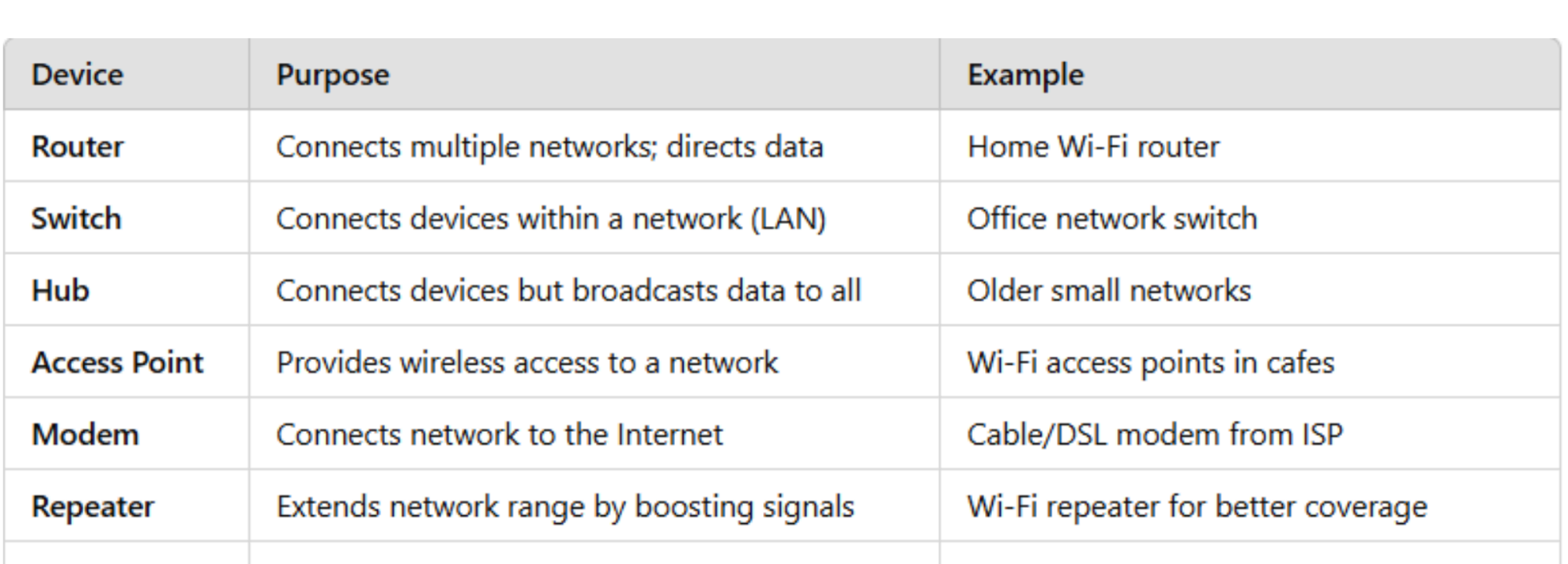
- **Function**: A repeater strengthens or amplifies the signal in a network, allowing it to travel further without degradation.
- **How It Works**: A repeater receives a weak or low-quality signal, amplifies it, and retransmits it at a higher strength. It doesn’t alter the data, just boosts the signal to extend the range.
- **Usage**: Used in large buildings, long cable runs, or Wi-Fi dead zones to ensure the signal reaches all areas.
- **Example**: In a large house with poor Wi-Fi in certain rooms, a Wi-Fi repeater can extend the range of the existing signal to improve connectivity.

9. Bridge

- **Function**: A bridge connects and divides networks to reduce traffic and manage network resources effectively. Bridge operates at data link layer
- **How It Works**: Bridges connect different segments of a LAN, forwarding data only if it’s intended for a device on the other side. They help divide traffic, making networks faster and more efficient.
- **Usage**: Used to improve network performance and create larger networks by connecting smaller segments.
- **Example**: In a building with multiple floors, a bridge can connect the networks on each floor, ensuring efficient data flow without mixing all traffic.

10. Gateway

- **Function**: A gateway serves as an entry and exit point for data between two networks that use different protocols (rules for data exchange). It translates data between networks so that devices can communicate.
- **Usage**: Gateways are used to connect different types of networks, like a company’s internal network with the Internet, or two different companies’ networks.
- **Example**: When a company network connects with the Internet, a gateway translates the internal network’s data protocols to those compatible with the Internet.



Device	Purpose	Example
Router	Connects multiple networks; directs data	Home Wi-Fi router
Switch	Connects devices within a network (LAN)	Office network switch
Hub	Connects devices but broadcasts data to all	Older small networks
Access Point	Provides wireless access to a network	Wi-Fi access points in cafes
Modem	Connects network to the Internet	Cable/DSL modem from ISP
Repeater	Extends network range by boosting signals	Wi-Fi repeater for better coverage
Bridge	Connects two similar networks	Connecting networks in an office
Gateway	Connects networks with different protocols	Company network gateway to Internet

1. Different Protocols in CN

Application Layer Protocols

1. **HTTP (Hypertext Transfer Protocol)**
    - **Overview**: A protocol for transferring hypertext (web pages) over the internet.
    - **Use**: Browsing websites (e.g., loading pages on Chrome or Firefox).
    - **Example**: A user enters a URL in their browser, and an HTTP request is sent to retrieve the web page.
2. **FTP (File Transfer Protocol)**
    - **Overview**: Used to transfer files between computers on a network.
    - **Use**: Uploading and downloading files from web servers.
    - **Example**: A website developer uses FTP to upload files from their local computer to their web server.
3. **SMTP (Simple Mail Transfer Protocol)**
    - **Overview**: A protocol for sending emails from clients to servers.
    - **Use**: Sending emails.
    - **Example**: When a user sends an email from a client like Outlook, SMTP is used to deliver the message to the recipient's mail server.
4. **POP3 (Post Office Protocol 3)**
    - **Overview**: A protocol for retrieving emails from a mail server.
    - **Use**: Downloading emails to a local client.
    - **Example**: A user uses an email client like Thunderbird to check their emails, which retrieves messages via POP3.
5. **IMAP (Internet Message Access Protocol)**
    - **Overview**: A protocol for accessing and managing emails on a mail server.
    - **Use**: Synchronizing email across multiple devices.
    - **Example**: A user accesses their email account from a phone and a laptop using IMAP, with changes reflected on both devices.
6. **DNS (Domain Name System)**
    - **Overview**: Translates domain names into IP addresses.
    - **Use**: Resolving user-friendly domain names to numerical IP addresses for locating web resources.
    - **Example**: When a user types www.example.com, DNS translates it to the corresponding IP address for the web server.
7. **SSH (Secure Shell)**
    - **Overview**: A protocol providing secure remote access to computer systems.
    - **Use**: Securely managing remote servers.
    - **Example**: A system administrator uses SSH to log into a remote Linux server to perform maintenance tasks.
8. **RTP (Real-time Transport Protocol)**
    - **Overview**: A protocol for delivering audio and video over IP networks.
    - **Use**: Used in streaming media and real-time communication systems.
    - **Example**: Streaming a live sports event or video conference call.

Transport Layer Protocols

1. **TCP (Transmission Control Protocol)**
    - **Overview**: A reliable, connection-oriented protocol that ensures accurate data delivery.
    - **Use**: Applications requiring data integrity, such as file transfers and web browsing.
    - **Example**: When a user downloads a file, TCP manages packet delivery to ensure the file is complete and intact.
2. **UDP (User Datagram Protocol)**
    - **Overview**: An unreliable, connectionless protocol prioritizing speed over reliability.
    - **Use**: Applications where speed is critical, such as streaming or gaming.
    - **Example**: Online gaming often uses UDP to quickly transmit player actions without waiting for acknowledgment.

Network Layer Protocols

1. **IP (Internet Protocol)**
    - **Overview**: Handles addressing and routing of packets across networks.
    - **Use**: Assigning unique IP addresses to devices and directing packets to their destinations.
    - **Example**: Data packets are sent from a user's device to a web server using IP addresses for routing.
2. **ICMP (Internet Control Message Protocol)**
    - **Overview**: Used for error reporting and diagnostic messages.
    - **Use**: Managing network conditions and reporting errors.
    - **Example**: The ping command uses ICMP to check the availability of a device on the network.
3. **ARP (Address Resolution Protocol)**
    - **Overview**: Maps IP addresses to physical MAC addresses.
    - **Use**: Facilitating communication within a local network by determining the hardware address associated with an IP address